

NN Optimization Strategies

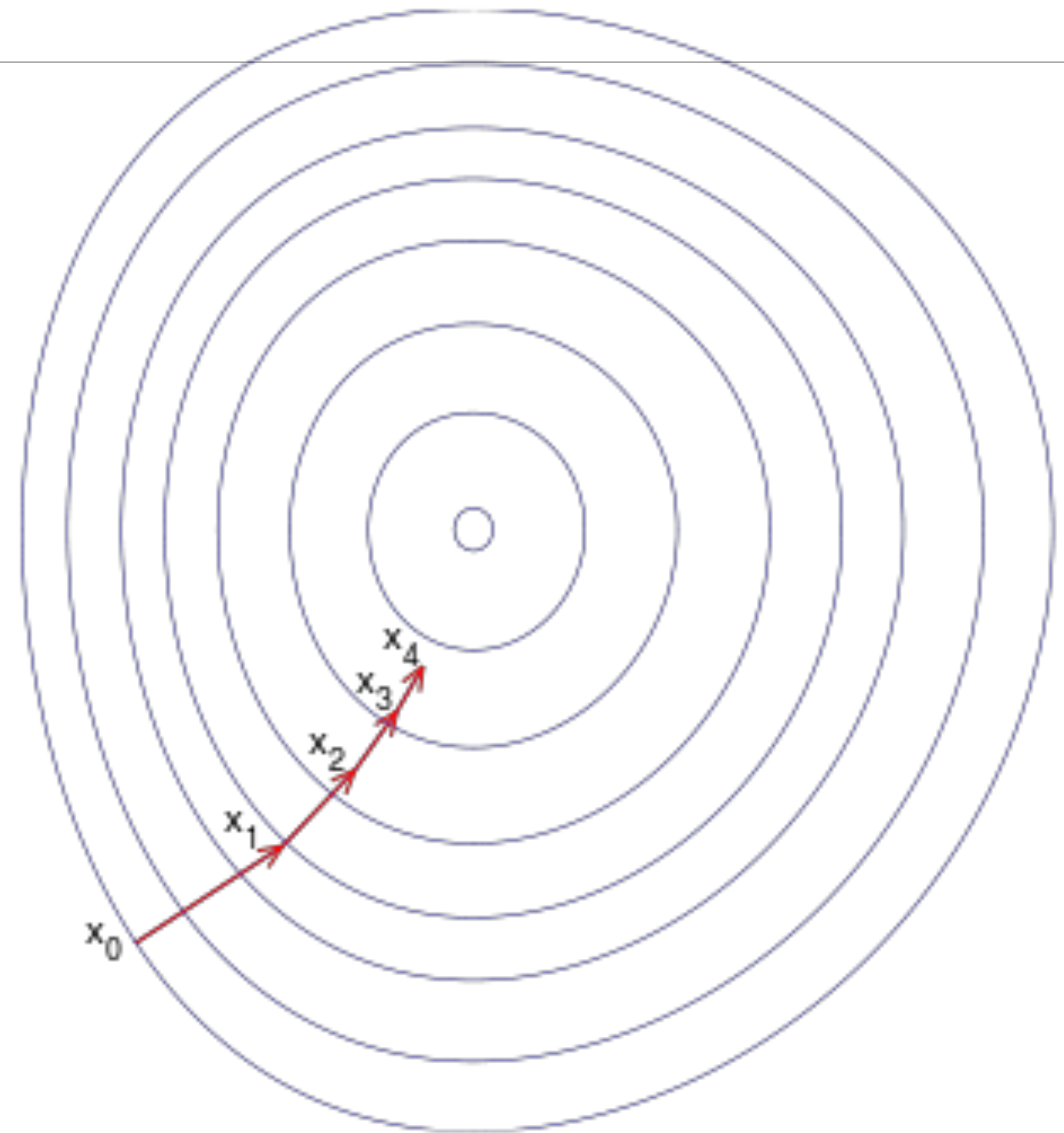
CSC 487: Deep Learning
Instructor: Jonathan Ventura

Minimizing the Loss Function

- Gradient descent:

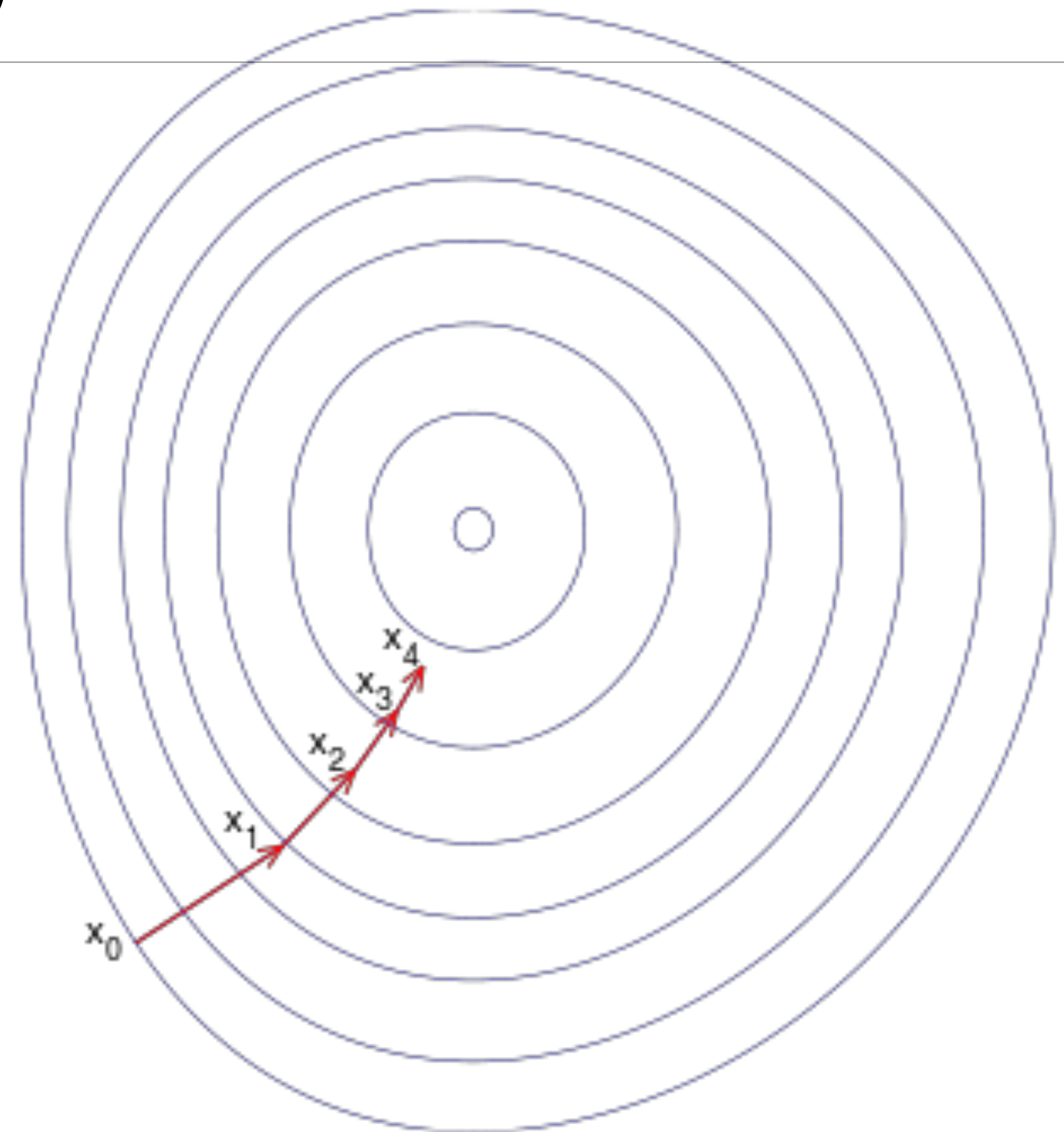
$$\mathbf{w} = \mathbf{w} - s \cdot \nabla \sum_i L_i(\mathbf{w})$$

- Problem?
 - Computing loss over entire dataset might be very expensive
 - We need to repeat this thousands of times

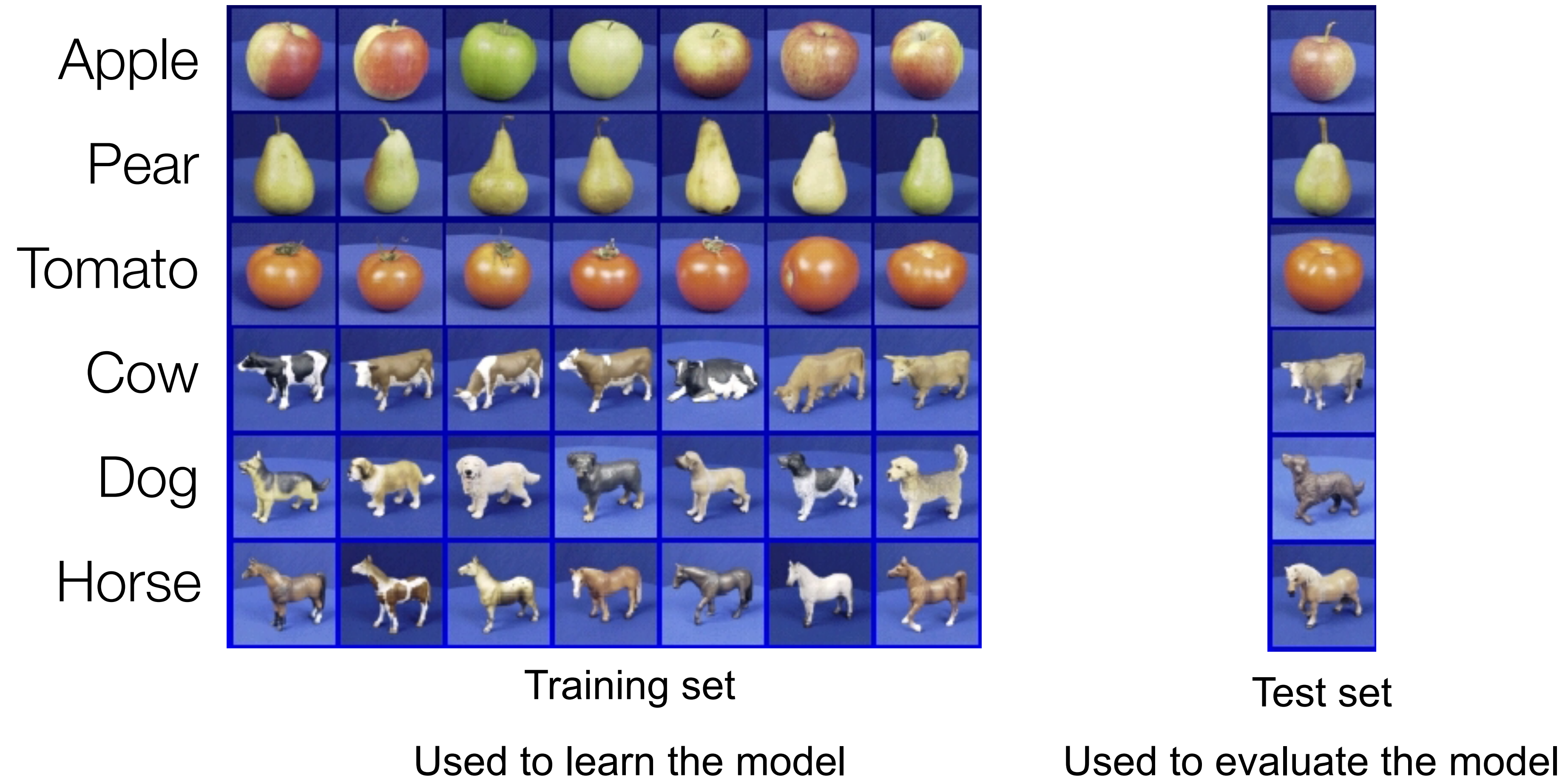


Stochastic Gradient Descent (SGD)

- Compute loss and gradient using a random subset of the dataset
 - **Batch** = random subset
 - Could use batch size of 32, or 256, ...
- Approximates the actual gradient vector
 - Larger batch = more accurate gradient
- **Epoch** = enough batches to see entire dataset once



Training and Testing Sets



Generalization



Training set



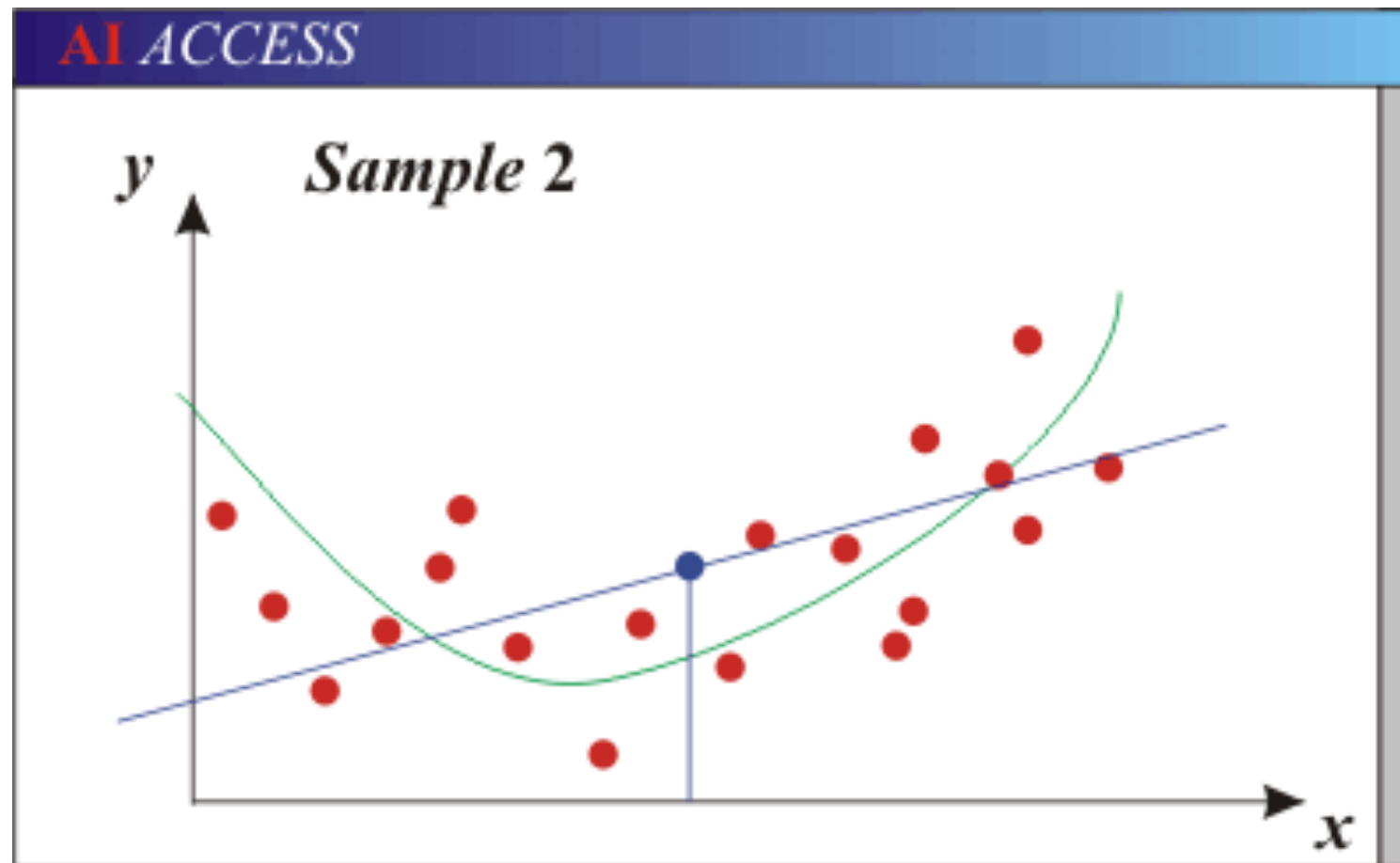
Test set

- How well does a learned model generalize from the data it was trained on to a new test set?

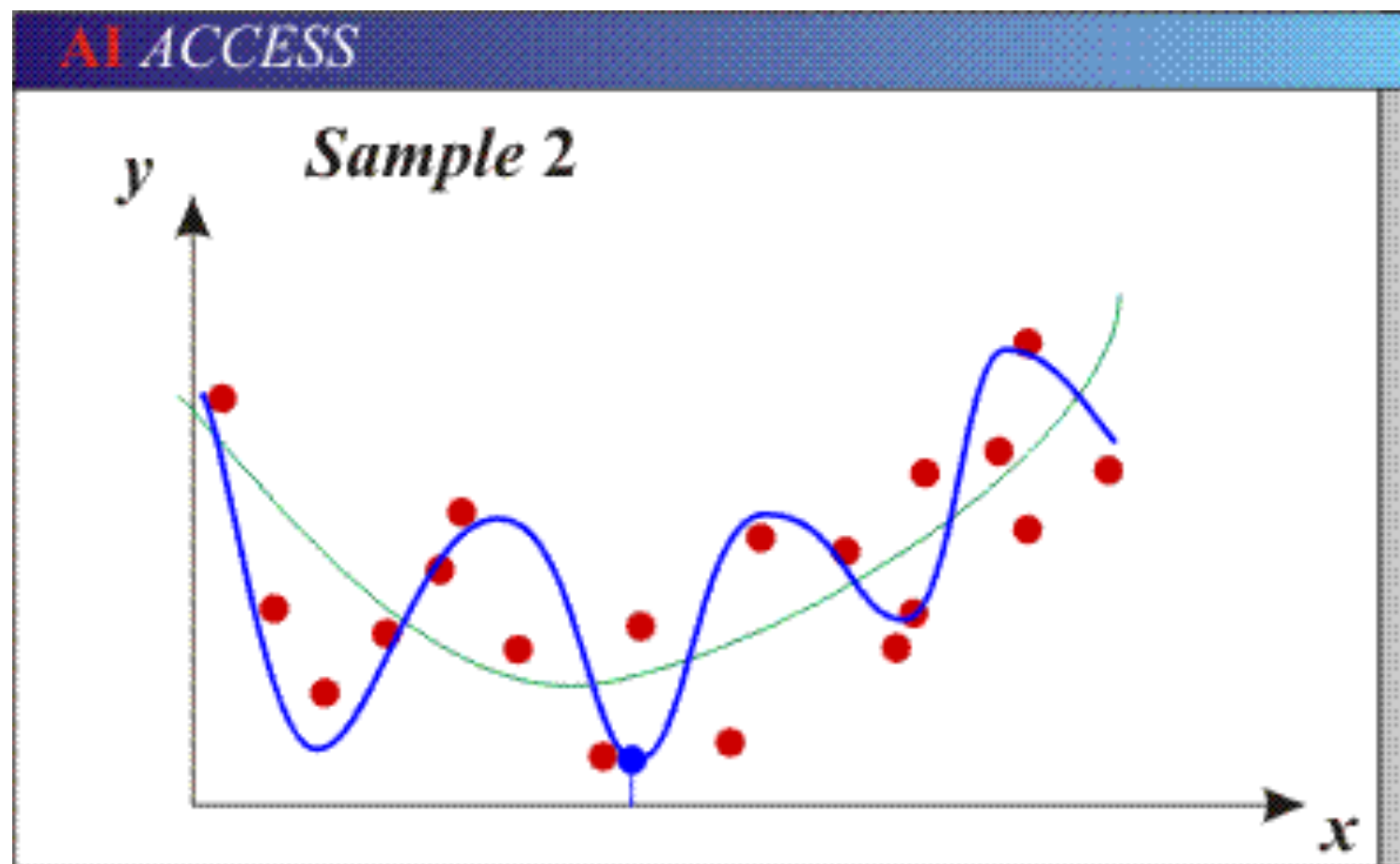
Generalization

- Components of generalization error
 - **Bias:** how much the average model over all training sets differ from the true model?
 - Error due to inaccurate assumptions/simplifications made by the model
 - **Variance:** how much models estimated from different training sets differ from each other
- **Underfitting:** model is too “simple” to represent all the relevant class characteristics
 - High bias and low variance
 - High training error and high test error
- **Overfitting:** model is too “complex” and fits irrelevant characteristics (noise) in the data
 - Low bias and high variance
 - Low training error and high test error

Bias-Variance Trade-off



- Models with too few parameters are inaccurate because of a large bias
 - Not enough sensitivity to the sample



- Models with too many parameters are inaccurate because of a large variance
 - Too much sensitivity to the sample

How to reduce variance?

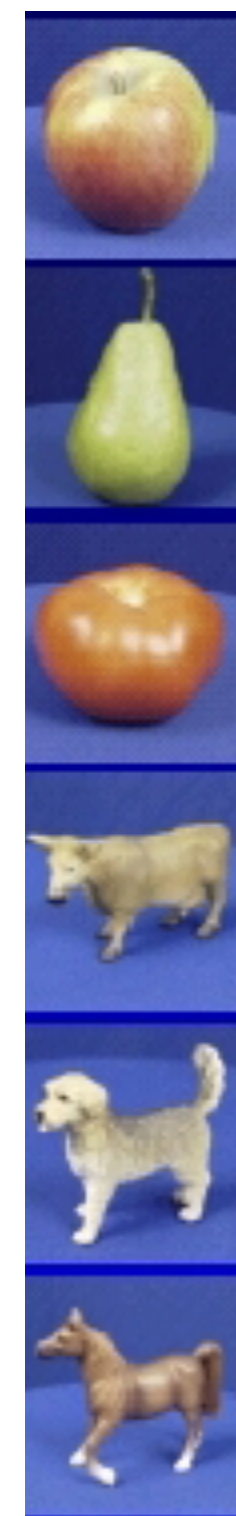
- Choose a simpler classifier
- Regularize the parameters
- Get more training data

Validation set

- Split training into “training” and “validation”
- Use validation for hyper-parameter tuning



Training set



Validation set

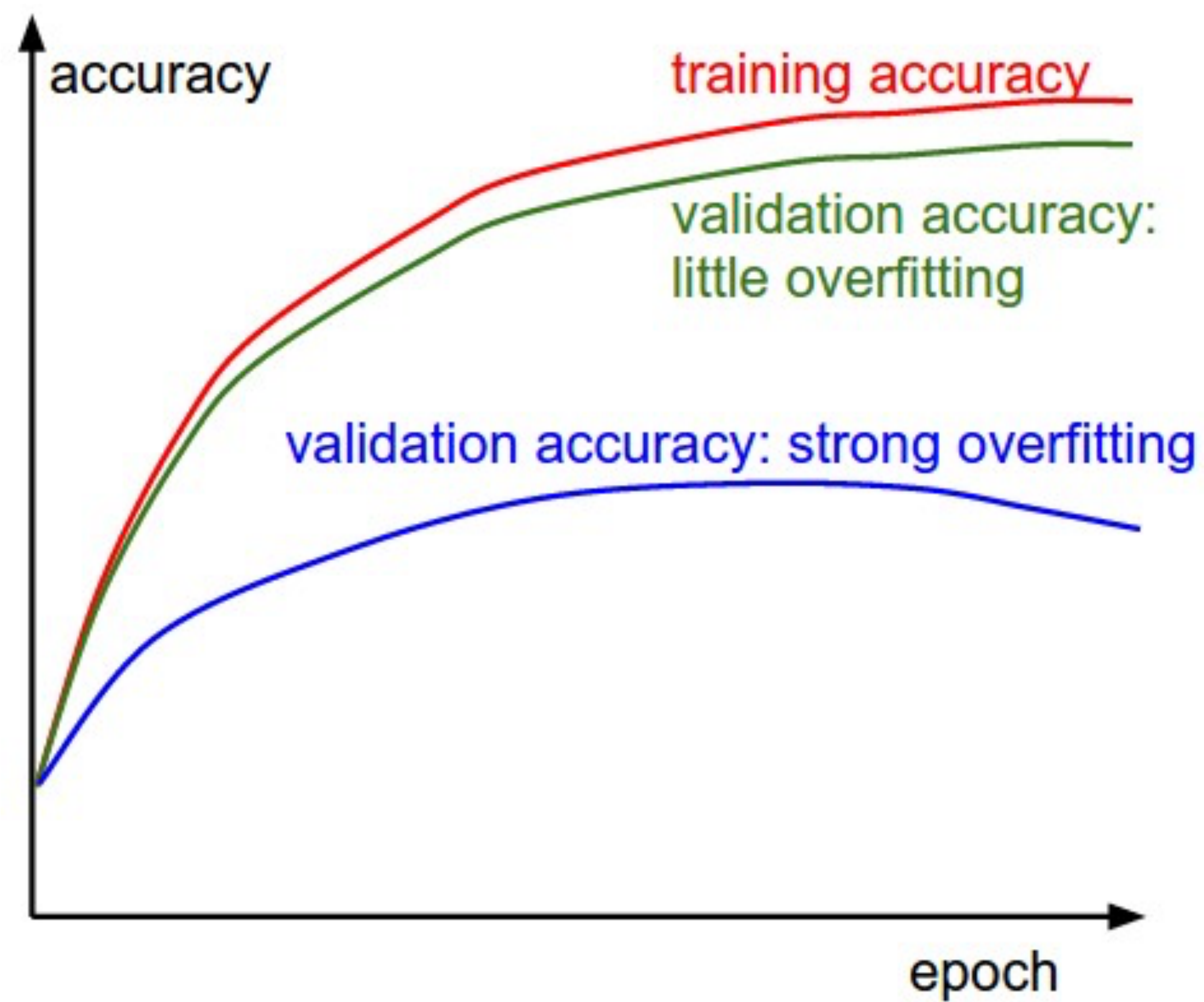


Test set

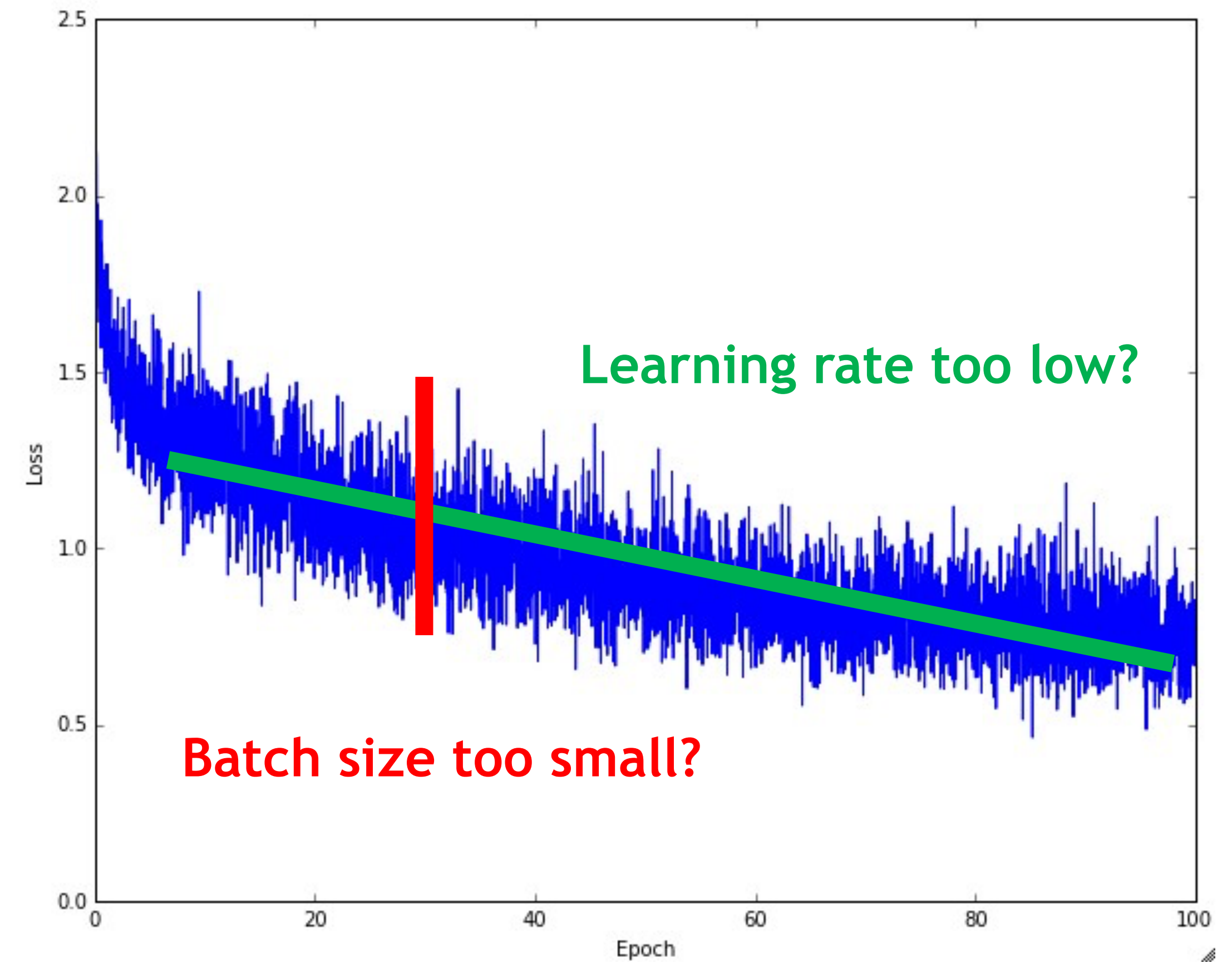
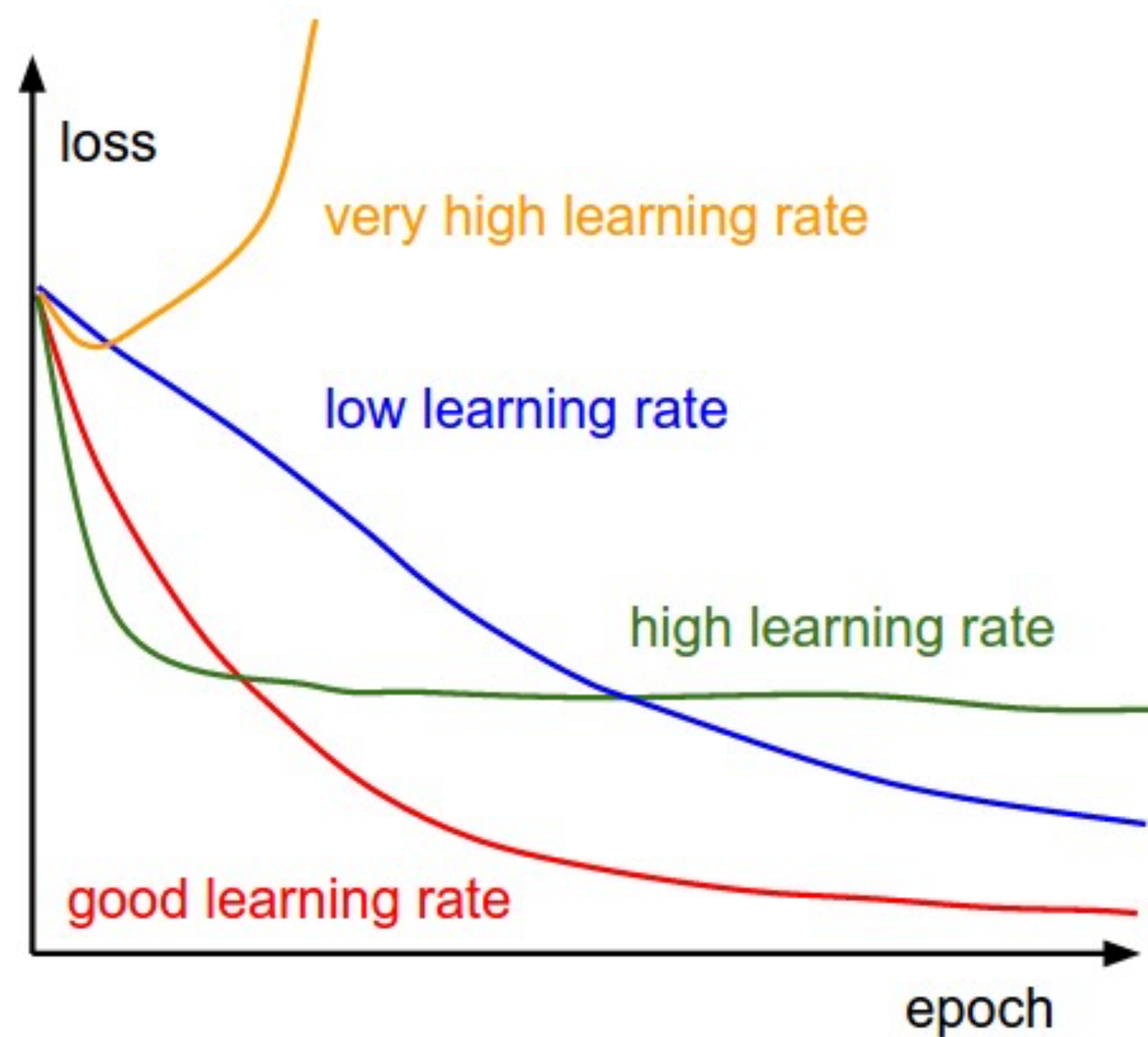
Hyperparameters

- Any parameter that is not learned in the training process
- Examples:
 - Network architecture (number of layers and channels)
 - Learning rate
 - Number of epochs
 - Batch size

Train/Val accuracy



Choice of learning rate and batch size



Stochastic Gradient Descent

- Basic gradient descent:

$$w_j \leftarrow w_j - s \cdot \frac{\partial L}{\partial w_j}$$

- Issue:
 - Gradients might be inaccurate due to batching in stochastic gradient descent
 - Might take a step in the wrong direction
 - Need to set learning rate very low to ensure stable improvement of loss function

Stochastic Gradient Descent with Momentum

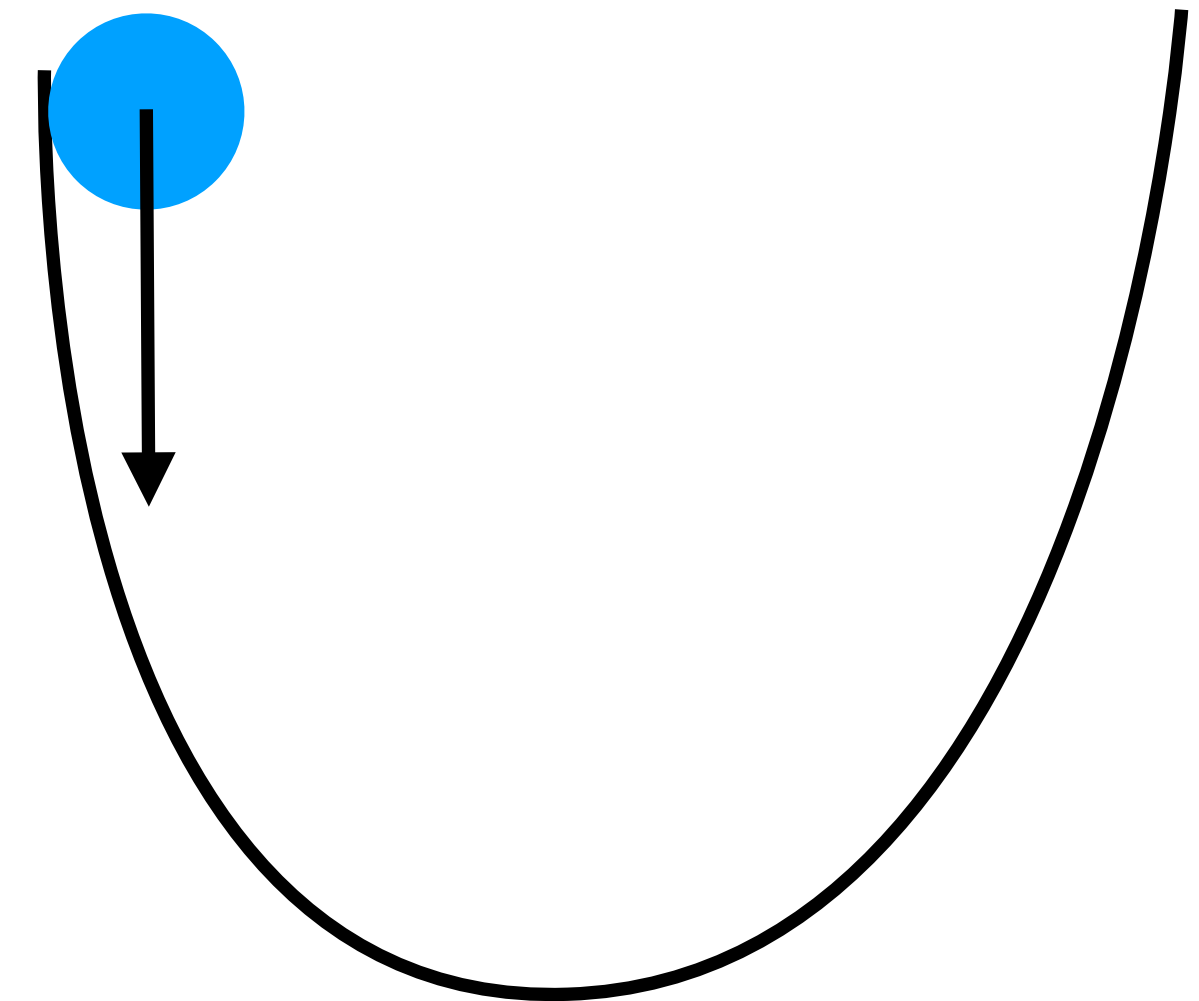
- Idea: maintain estimate of “velocity” and keep the ball rolling down the hill:
 - Update velocity with damping factor β_1 (common choice is $\beta_1 = 0.9$)
 - Gradient represents “force”

- Velocity update:

$$m_j \leftarrow \beta_1 \cdot m_j + (1 - \beta_1) \cdot \frac{\partial L}{\partial w_j}$$

- Weight update:

$$w_j \leftarrow w_j - s \cdot m_j$$



Adaptive Moment Estimation (Adam)

Kingma and Ba, 2014

- Idea: also maintain estimate of variance of gradient

- First moment update:

$$m_j \leftarrow \beta_1 \cdot m_j + (1 - \beta_1) \cdot \frac{\partial L}{\partial w_j}$$

- Second moment update:

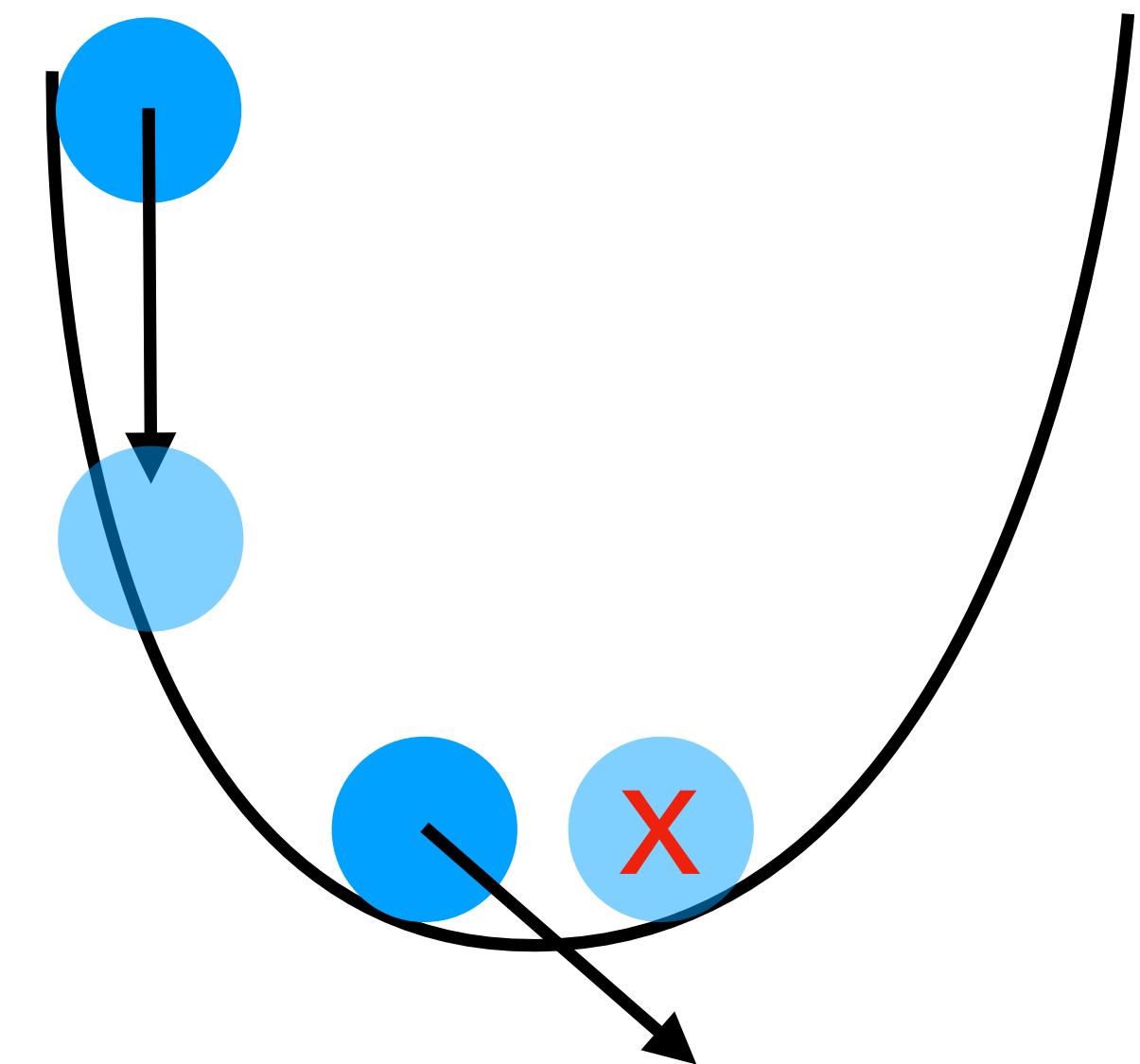
$$v_j \leftarrow \beta_2 \cdot v_j + (1 - \beta_2) \cdot \left(\frac{\partial L}{\partial w_j} \right)^2$$

- Weight update:

$$w_j \leftarrow w_j - s \frac{m_j}{\sqrt{v_t} + \epsilon}$$

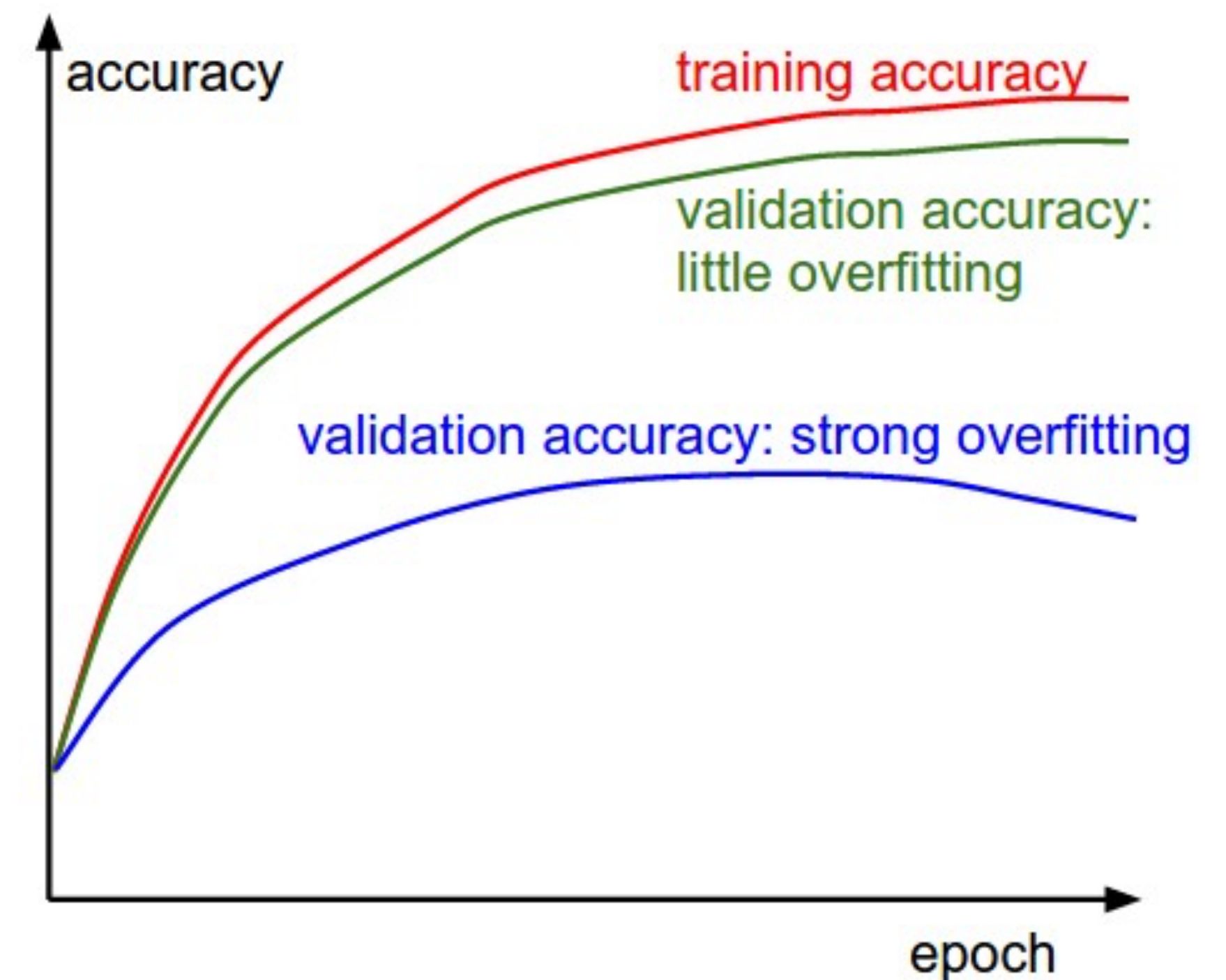
Annealing the Learning Rate

- Issue: as we approach minimum, need to take smaller steps to avoid “overstepping” the minimum
- Idea: Reduce the learning rate over time
 - Can multiply by λ after every epoch:
 $s \leftarrow s \cdot \lambda$
 - Or, reduce learning rate when validation accuracy doesn't improve for several epochs



Stopping Criterion

- Don't use number of epochs as a hyperparameter
 - Instead, adaptively stop training when overfitting is detected.
- Stop training when validation loss or accuracy doesn't improve for some number of epochs



Summary

- General machine learning process:
 - Split data into training, validation and testing sets
 - Fit model to training data (stochastic gradient descent)
 - Use validation data to compare competing models, determine choice of hyperparameters, check for overfitting
 - Evaluate final model on test data